

Ernest Wong

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EDUCATION

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| Simon Fraser University , Burnaby, BC
<i>Bachelor of Science in Computer Science</i> | December 2027 |
| University of Glasgow , Scotland, UK
<i>Field Studies in Cybercrime</i> | August 2024 |
| British Columbia Institute of Technology , Vancouver, BC
<i>Diploma in Cloud Computing Systems, Completed with Distinction</i> | December 2021 |

TECHNICAL EXPERIENCE

Oakmont Industrial <i>Software Engineer Co-op</i>	September 2026 - Present
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- Unblocked an ERP integration proof of concept by validating Aquilon's XML and JSON import path over SFTP and confirming that production jobs must originate inside the ERP.
- Mapped eMullion fields to Aquilon's sales-order schema and designed a Go transfer service to eliminate duplicate order entry.

Elections Canada Government of Canada <i>Data Analyst Co-op</i>	May 2026 - Present
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- Automated monthly EAP reporting with a Python pipeline that converts multi-page Tableau PDFs into Power BI-ready CSVs, replacing manual transcription and recovering metrics omitted from earlier datasets.
- Made the pipeline resilient to new reporting years by splitting extraction into bilingual parsers for five dashboard sections and validating decimals, missing fields, and zero-versus-null values.
- Enabled non-technical analysts to calculate issue rates without SQL by moving recurring business rules into reusable Denodo views consumed by Power BI.

ISED Government of Canada <i>Site Reliability Engineer Cloud Ops Student Programmer Co-op</i>	January 2025 - August 2026
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- Led releases for a \$100M+ patent platform across AWS, Salesforce, and Java teams, restoring services within approved maintenance windows after application freeze and validation.
- Automated configuration drift detection across 8+ environments with Java, GraphQL, and Jenkins, replacing manual comparisons with daily mismatch reports.
- Automated CI/CD configuration for 50+ applications with Python and Selenium across Bitbucket, Bamboo, Nexus, and IBM WebSphere, replacing repeated manual setup.
- Resolved UI/UX and API defects in Salesforce Lightning components affecting bilingual submissions, improving reliability for high-volume online patent filings.
- Co-designed a Duolingo-inspired equity, diversity, and inclusion (EDI) training module in under 6 hours, earning finalist honors for “Most Impactful” and “Best Use of Google Gemini” at Canada’s largest AI hackathon and featured in the ISED newsletter.

UBC Orbit | SFU SAT Engineering Design Team

September 2023 - September 2026

Executive | Mission Operations Developer | Embedded Systems Lead

- Led a high-altitude balloon mission through design and flight-readiness reviews, integrating Python software with 10+ Raspberry Pi sensors before a flight to 30 km.
- Built ALEASAT's React mission-operations dashboard to visualize telemetry and execute command workflows for a spacecraft planned for a European Space Agency mission and 2028 SpaceX launch.
- Migrated outreach websites from Squarespace to GitHub Pages and React, eliminating hosting costs.

École Montgomery Middle School

September 2024 – December 2024

After-school Computer Instructor

- Designed and delivered coding lessons for a class of 20 students using Minecraft Education, block-based programming, and Python.
- Introduced web development through CodePen and GitHub Pages, enabling students to design and publish their own websites.
- Adapted technical concepts into accessible, hands-on activities that encouraged experimentation, creativity, and independent problem-solving.

Simon Fraser University | Faculty of Applied Sciences

May 2024 - August 2024

Research Assistant in AI Cybersecurity

- Developed a machine learning tool using Python to remove personal information from AI models achieving F1 greater than 80% on evaluation datasets.
- Identified potential security risks in AI models, such as unauthorized access to sensitive data, using Tensor Trust to demonstrate vulnerabilities.

LEADERSHIP & TEACHING

Simon Fraser University | Faculty of Applied Sciences

January 2024 - Present

Student Ambassador | Computing Science Peer Tutor

- Delivered engaging presentations at university recruitment events, reaching 100+ prospective students, inspiring two individuals to pursue Computer Science at Simon Fraser University and join SFU SAT.
- Developed an automated system using Python and Selenium to efficiently send 200 personalized emails to schedule mentoring sessions within a day.
- Engaged students as a computer peer tutor in explorative discussions, encouraging curiosity in introductory programming concepts in Python.
- Collaborated closely with instructors to curate personalized three-hour one-on-one tutoring sessions.

CERTIFICATIONS & AWARDS

- AWS Certified AI Practitioner
- Amateur Radio Operator Certificate
- Computer Science Undergraduate Society Award (one of five awarded annually for SFU contributions)
- Recognition Award from the Mayor of Richmond for Community Contributions
- Global Skills Opportunity: Equity, Diversity, and Inclusion International Mobility Award

TECHNICAL SKILLS

Languages: Python, Go, Java, TypeScript, JavaScript, SQL, Bash

Systems: React, Node.js, GraphQL, REST, Linux, AWS, Docker, Kubernetes, OpenShift, Salesforce

Data and Delivery: Pandas, Denodo, Power BI, Jenkins, GitHub Actions, Selenium, JUnit, Pytest, Git